

Chapter

Reframing Quality Management: Drivers for the Development of Quality 5.0

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Abstract

Quality management has undergone significant advances and transformations driven by digital technology, leading to the emergence of Quality 5.0 (Q5.0). In this sense, organizations require comprehensive guidance to adapt their processes and incorporate more human-centric, sustainable, and resilient values. This chapter aims to propose drivers for the development and improvement of Q5.0, with the objective of supporting the modernization of quality management within organizations in light of Industry 5.0. To develop this proposal, a content analysis of relevant, recent, and reliable scientific literature on this topic was conducted. This analysis enabled the identification of boosting elements that underpin the successful implementation of Q5.0 in industrial organizations. These elements were subsequently grouped by affinity, which yielded the proposed drivers designed to support managers in implementing and consolidating Q5.0 within their organizations.

Keywords: Quality 5.0, Industry 5.0, quality management, smart technologies, smart systems

1. Introduction: From traditional quality to Quality 5.0

The evolution of quality in organizations stems primarily from significant societal changes. Technological advances and geopolitical issues require organizations to continuously improve how they define and deliver quality processes, products, and services. In this regard, it is possible to identify temporal milestones that delineate distinct periods, each characterized by different focuses and approaches to quality. Initially, quality was limited to inspecting finished products and classifying them as “good” or “defective” (Quality 1.0). With the adoption of statistical tools for process control and monitoring (Quality 2.0), organizations began to focus on defect detection and its relationship to variables such as raw materials, employees, and equipment. In the post–World War II period, and in response to new demands and consumption patterns, several professionals in the field continued to seek and

develop new approaches, methods, and tools to improve the quality of processes, products, and services. These efforts resulted in the consolidation of total quality management (Quality 3.0), in which the concept of quality was expanded toward a more systemic, participatory perspective, oriented toward problem prevention and continuous improvement, with a customer focus and consideration of external factors [1–3]. **Figure 1** summarizes the main milestones in the evolution of quality management from Quality 1.0 to Quality 5.0.

With technological advances in integrating digital technologies into production systems, quality evolved toward data-driven models featuring real-time monitoring and predictive capabilities, while also supporting more flexible, customization-oriented production environments [4]. As organizations systematically incorporated these technologies into quality management practices, Quality 4.0 (Q4.0) emerged as a consolidated concept in the literature. In this context, Industry 4.0–related technologies, such as sensors, the Internet of Things (IoT), and cyberphysical systems, enable continuous data collection, supporting real-time monitoring and the early detection of deviations and defects. At the same time, the use of big data (BD) and artificial intelligence (AI) have increased the volume and velocity of available data, enabling the conversion of data into useful information for quality management and more agile decision-making [5]. This need for rapid responses arises from increasingly dynamic markets that require shorter lead times and customized production. In this sense, Q4.0 enables more strategic management by integrating production, quality, maintenance, and supply chain data, providing a systemic view and enabling data-driven decision-making [6, 7].

However, recent transformations in industry have expanded expectations of organizations' roles, meaning that efficiency and economic performance alone are no longer sufficient. In this context, although relevant, the digital advances associated with Q4.0 tend to emphasize technological aspects and operational efficiency, often without adequate attention to human and social value.

To address new social, market, and stakeholder demands and align with the emerging concept of Industry 5.0 (I5.0), Quality 5.0 (Q5.0) has emerged. This new approach reorients quality management by prioritizing three main elements: human-centricity, sustainability, and resilience, while maintaining the pursuit of efficiency through technological resources [8]. In Q5.0, human-centricity assumes a central role in production processes. By combining human skills with robot capabilities, operations can be performed more quickly while also providing better ergonomics and working conditions [9]. Sustainability also gains greater relevance in production processes through the adoption of practices aligned with the circular economy and resource-use optimization. This approach includes real-time

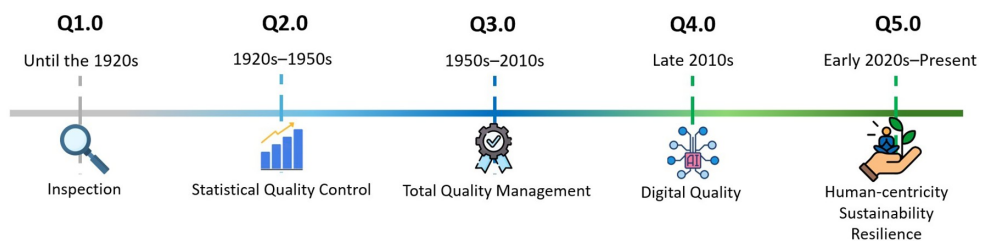


Figure 1.
Quality evolution. Source: The authors.

monitoring of energy, water, and raw material consumption, as well as the development of environmentally responsible products. Such practices enhance operational efficiency while reducing environmental impacts. Resilience, in turn, is primarily the capacity to adapt to crises arising from adverse events, such as economic instability, climate change, and wars [10].

Therefore, Q5.0 can be defined as the fifth generation of quality management, a new paradigm that expands on previous concepts by emphasizing human-centricity and social well-being, sustainability, and resilience aligned with technological development [11]. Thus, Q5.0 does not replace previously practiced management models but broadens the boundaries of quality by highlighting that sustainability and human well-being are inseparable from management practices. However, Q5.0 may be challenging for many companies, which may face barriers in reconciling technology implementation with employees' adaptation to new work demands, as well as in aligning processes with environmental targets [12]. Despite these challenges, they can be overcome through a structured, strategic approach, supported by studies that propose models, guidelines, and boosters for the transition to Q5.0. These challenges do not represent permanent barriers but rather transitional stages in an organizational transformation process that demands planning, collective learning, and institutional commitment.

Given the potential for further evolution in quality models within organizations, this chapter aims to support the development of Q5.0 by proposing drivers that guide quality management toward models aligned with I5.0. These drivers comprise guidelines that companies should adopt to automate and digitalize their processes without losing sight of the social, environmental, and economic dimensions, thereby enabling a more systemic, strategy-aligned approach to management. This proposal began with a systematic literature review, followed by an in-depth content analysis of the selected studies. This process enabled the identification of the factors that contribute to and enable the development and improvement of Q5.0, which were termed boosting elements. These elements were synthesized and grouped by their thematic affinities, yielding three drivers: environmental sustainability and resilience; smart systems and human-machine collaboration; and social responsibility and quality of work life. From a practical standpoint, these drivers serve as a reference for prioritizing initiatives, assisting in decision-making, and structuring implementation roadmaps. From a theoretical standpoint, they help organize scattered studies and advance the state of the art by proposing a framework that can support future operationalizations, metrics, and empirical validations in the field of Q5.0.

This chapter is structured into five sections. Following this introduction, Section 2 describes the research method. Section 3 presents the Q5.0-boosting elements identified in the literature, while Section 4 proposes and discusses the drivers for the development and improvement of Q5.0. Finally, Section 5 summarizes the main conclusions, highlights the theoretical and practical contributions of the chapter, and outlines directions for future research.

2. Research method

This study employed the systematic literature review (SLR) research method, which advances knowledge in a field of study by analyzing scientific documents,

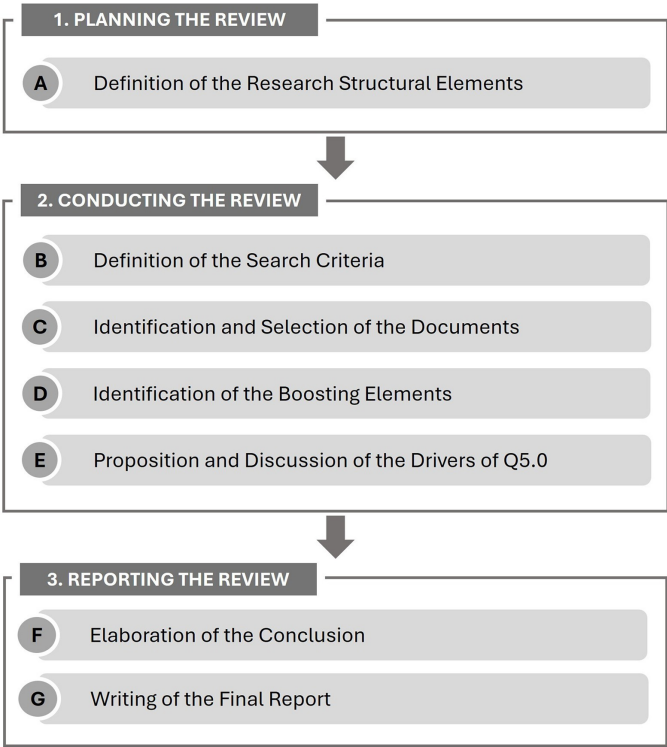


Figure 2.
Methodological flow. Source: The authors.

such as articles, reviews, book chapters, and other relevant sources. The SLR enables in-depth exploration and understanding of a topic, the identification of scientific gaps, and the synthesis of existing knowledge [13]. This method is often structured in three main phases: (1) planning the review; (2) conducting the review; and (3) reporting the review [14]. These phases were applied and adapted to the study objectives, as shown in **Figure 2**.

Figure 2 illustrates the methodological flow of this study, which covers three main phases and seven subphases. Each phase is detailed in Sections 2.1, 2.2, and 2.3.

2.1 Planning the review

In this phase, the structural elements of the research, such as the theme, scientific gap, objective, and research method, were defined (Subphase A).

2.2 Conducting the review

The objective of this phase was to identify, select, and analyze scientific documents related to Q5.0. To this end, the search criteria used in the SLR (Subphase B) were defined, as shown in **Figure 3**.

As shown in **Figure 3**, the Scopus and Web of Science databases were selected to retrieve scientific documents. They were chosen due to their broad coverage and the

 DATABASES	• Scopus • Web of Science
 SEARCH FIELDS	OR Title, abstract or keywords "Quality 5.0" Title Quality and "Industry 5.0"
 DOCUMENT TYPE	• Articles • Reviews
 LANGUAGE	• English

Figure 3.
Search criteria. Source: The authors.

quality of the journals they index in this research field [15–17]. The term “Quality 5.0” was searched for in the titles, abstracts, and keywords of the indexed documents. However, some studies address quality management in the context of Industry 5.0 without explicitly using the term “Quality 5.0.” To identify these studies, the search strategy was expanded by including the terms “quality” and “Industry 5.0.” These additional terms were searched only in the document titles to retrieve studies directly related to the topic of this research. The search was limited to English-language articles and reviews. Only articles and scientific reviews were selected, as they are more reliable documents and undergo peer review [18]. The restriction to English is due to its predominance in academia [13].

In Subphase C, the documents for the SLR were identified and selected from the databases used. To make this process more transparent, reproducible, and rigorous, the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram was employed (**Figure 4**).

As illustrated in **Figure 4**, 62 documents were identified in Scopus and 33 in the Web of Science database in January 2026 using the search terms presented in **Figure 3**. The automatic filters available in these databases were used to narrow the document type and language, resulting in the exclusion of 28 records. After merging the records retrieved from both databases, 25 duplicate documents were identified and removed.

The remaining 42 documents were then screened based on their titles, abstracts, and keywords. Full-text reading was performed only when this information was insufficient to determine the study’s eligibility. As a result, nine documents were excluded because they did not specifically address Quality 5.0 and were therefore considered outside the scope of this study. In addition, three documents were excluded because their full texts were unavailable. Finally, 30 scientific documents were selected to compose the SLR.

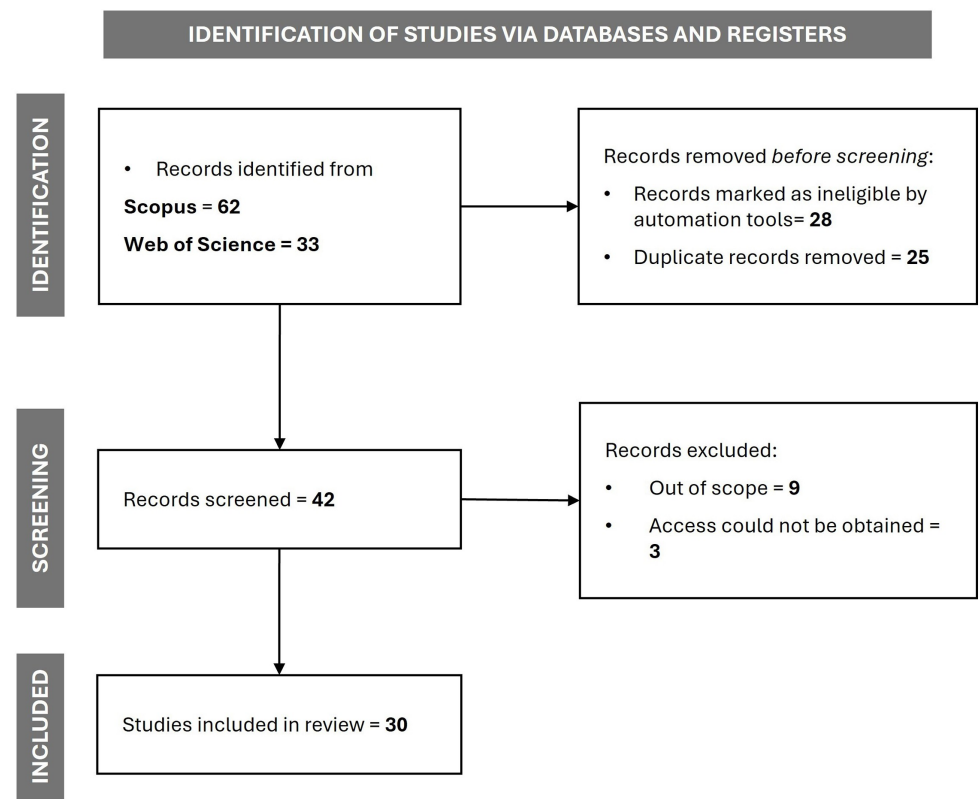


Figure 4.
PRISMA flow diagram. Source: The authors.

Subsequently, the process of analyzing these documents began to identify the boosting elements of Q 5.0 (Subphase D). To this end, the content analysis research method was used, which enables the interpretation and organization of textual information through systematic, in-depth reading. Thus, based on the scientific literature and the authors' expertise, it was possible to identify elements that guide and provide strategies for promoting Q 5.0, referred to as boosting elements. These elements are presented in Section 3.

In Subphase E, the boosting elements were grouped by their thematic affinity and clustered into driver groups. This process resulted in the proposal of three drivers for the development and improvement of Q 5.0, namely environmental sustainability and resilience; smart systems and human-machine collaboration; and social responsibility and quality of work life. These drivers are presented and discussed in Section 4 of this chapter.

2.3 Reporting the review

In this phase, the main results of this study were summarized (Subphase F), highlighting the answer to the research question, filling the scientific gap, and achieving the established objectives. Theoretical and practical contributions are presented, along with suggestions for future research. Finally, the final draft of this

SLR was written, which involved organizing and discussing all the findings of this study (Subphase G).

3. Boosting elements of Q5.0

In this section, the 30 boosting elements of Q5.0 identified in the scientific literature are presented and classified into three dimensions of I5.0, as formalized in the European Commission's report on the future of industrial production [8]: *human-centricity* (H), *sustainability* (S), and *resilience* (R).

These boosting elements were classified into I5.0 dimensions based on evidence from the scientific literature and the authors' expert judgment, with consideration of each element's predominant focus. Although some elements may also relate to other I5.0 dimensions, only those reflecting their primary conceptual emphasis were assigned. This criterion was adopted to ensure conceptual consistency and avoid overly broad classifications based on secondary or indirect relationships.

The elements, their dimensions, and scientific references are presented in **Table 1**. The following paragraphs discuss the groups of elements associated with each dimension of I5.0 (H, S, and R), as well as those that encompass more than one dimension (H/S, H/R, and H/S/R).

Regarding the *human-centricity* (H) dimension, the identified elements are based on the premise that, from the company's internal perspective, quality management must align with employees' needs and use technology to support human judgment [8]. In this regard, elements that keep the human being as the central agent of control and decision-making stand out, using cyberphysical systems and advanced technologies to support knowledge, creativity, and quality-oriented decision-making [12, 24, 26, 27, 29–32, 35], while promoting innovation, efficiency, the appreciation of human capital, and the improvement of products and services [11, 30, 37, 39]. Human-machine collaboration is associated with safe interactions focused on employee well-being, satisfaction, trust, and commitment [24, 43], as well as real-time monitoring of safety conditions, ergonomics, and mental load, which positively impacts process quality [12, 22, 26, 41]. The use of digital platforms helps improve employees' experience and participation [11, 16, 22–24, 26, 29–31, 34, 37, 39] and, from an external perspective, stakeholders' experience and participation, enabling them to share information, provide feedback, and contribute to learning, health, and occupational safety [12, 21, 22, 24, 29–31, 40]. Finally, this dimension includes elements that value the incorporation of social indicators and the engagement of diverse stakeholders in defining quality policies and strategies, thereby reinforcing transparency, inclusion, and shared responsibility [30, 42].

In the context of the *sustainability* (S) dimension, Q5.0 incorporates sustainability as a performance criterion for quality management, in line with the I5.0 perspective [8]. In this regard, the identified elements indicate the use of I5.0 technologies to mitigate the environmental impacts of industrial operations, especially by reducing waste and improving energy efficiency [24, 26, 37]. Additionally, it emphasizes the importance of using environmental indicators to track and assess the organization's performance in environmental quality management, thereby enabling better decision-making and improving sustainability over time [11].

#	Boosting elements of Q 5.0	Dimensions of I5.0	Articles
1	Develop continuous Q 5.0 training programs, supported by AI, IoT, and virtual reality, to personalize learning, strengthen human competencies, and enhance adaptability to technological changes and market demands, addressing topics such as social responsibility, ethics, and digital technologies	H/R	[5, 11, 12, 19–26]
2	Develop a quality management system aligned with I5.0 to improve working conditions for human beings	H/S/R	[12, 19, 24, 26–32]
3	Incorporate environmental sustainability into quality management, shifting the focus from just the customer to the satisfaction of society, aiming to create sustainable value for the organization	H/S	[11, 12, 22, 24, 26, 29–31, 33, 34]
4	Design cyberphysical systems that combine data from sensors, computational models, and experience to support decision-making in quality management, while maintaining humans as the central agents of control	H	[12, 24, 26, 27, 29, 31, 32, 35]
5	Implement technologies such as AI and BD to identify new customer demands and customize products and processes, ensuring quality, sustainability, and human and social value throughout the product lifecycle	H/S/R	[5, 20, 24, 25, 34, 36–38]
6	Use digital technologies and resources responsibly and ethically to promote Q 5.0, enhancing employee experience and reducing the negative social impacts associated with technological innovation	H	[16, 22, 24, 26, 30, 31, 34, 39]
7	Develop management and leadership models that encourage employee coparticipation and collaboration in the digital environment, strengthening quality management through knowledge exchange, and the sharing of successful practices	H	[11, 23, 24, 29, 30, 37, 39]
8	Reorient quality management systems toward a human-centered perspective, integrating sustainability and risk management principles	H/S/R	[16, 20, 24, 30, 34, 39]
9	Develop corporate healthcare service models in organizations that use connected digital systems to strengthen quality management, enabling workers, healthcare professionals, and other stakeholders to share relevant data that support decision-making while promoting workforce health, learning, engagement, and well-being	H	[12, 21, 22, 29, 30, 40]
10	Structure resilient and risk-based supply chains to strengthen quality management and organizational recovery in crisis scenarios	R	[11, 23, 26, 29, 36]
11	Use digital platforms to receive feedback and opinions from employees and stakeholders, thereby promoting improvements in production processes, organizational culture, and occupational health and safety	H	[22, 24, 29–31]
12	Gradually develop an organizational culture aligned with Q 5.0 through pilot projects and structured change management actions, considering the technical and technological limitations of each organization	H/R	[19, 20, 24, 26, 37]
13	Integrate data from automated processes across different sectors into an interoperable platform to support quality management and informed human decision-making	H/R	[11, 24, 32, 34, 37]

#	Boosting elements of Q 5.0	Dimensions of I5.0	Articles
14	Drive technological innovation and process efficiency across the organization, prioritizing individual excellence, work-environment well-being, and the appreciation of human capital as a strategic factor in quality	H	[11, 30, 37, 39]
15	Implement robotic systems and functionalities for human-robot collaboration in quality management, integrating online monitoring of performance indicators and human-centered factors such as safety, mental load, physical and cognitive ergonomics, trust, and acceptance, to ensure safe, efficient interactions focused on employee well-being	H	[12, 22, 26, 41]
16	Use advanced technologies to support human knowledge, creativity and decision-making, and improve the quality of products and services	H	[12, 24, 30]
17	Predict and correct failures and defects throughout the production process and equipment lifecycle in real-time through human-machine collaboration	R	[19, 24, 32]
18	Implement I5.0 technologies to reduce waste and improve energy efficiency, mitigating negative environmental impacts	S	[24, 26, 37]
19	Integrate economic, environmental, and social data into the company's computerized quality management systems to support sustainable innovation and strategic decision-making	H/S	[30, 37, 42]
20	Expand the role of quality managers to act as agents of integration between organizational performance and social demands, in addition to being responsible for standards and audits	H/R	[30, 33, 37]
21	Use AI to configure robots serving customers so that, in addition to performing tasks efficiently, they also contribute to employee satisfaction, trust, and commitment, strengthening quality management in the context of Q 5.0	H	[24, 43]
22	Integrate digital systems for continuous training aimed at enhancing innovation and quality management in industrial environments through university-industry collaboration and investment in R&D	H/R	[37, 44]
23	Incorporate social indicators into digital quality management systems, such as gender diversity, equity in leadership positions, and working conditions, to monitor social impacts and align the organization with the Sustainable Development Goals (SDGs)	H	[30, 42]
24	Develop databases based on adaptive systems to support Q 5.0, balancing operational stability and organizational learning	H/R	[27, 36]
25	Create computerized predictive models using technologies that support problem-solving and quality management through adaptive decision-making and the overcoming of challenges in constantly changing production environments	R	[11, 24]
26	Plan and implement data-sharing through smart technologies across the supply chain to support Q 5.0, integrating active listening to consumers and other stakeholders	R	[5, 25]
27	Adopt and integrate internationally recognized management system models and standards, such as EFQM, ISO 9001, ISO 14001, and ISO 45001, to structure organizational processes and promote Q 5.0	H/S	[12]

#	Boosting elements of Q 5.0	Dimensions of I5.0	Articles
28	Use digital technologies to support the development and monitoring of indicators that enable the analysis of the organization's quality management performance in relation to sustainability	S	[11]
29	Develop digital platforms in a web environment with customizable options and suggested content based on user history, strengthening the quality of B2B (business-to-business) and B2C (business-to-consumer) relationships and promoting trust, transparency, and satisfaction	H/R	[38]
30	Involve stakeholders in creating policies, strategies, and decisions related to quality, aiming for transparency, shared responsibility, and inclusion	H	[30]

Table 1.

Boosting elements of Q 5.0 classified according to the dimensions of I5.0. Source: The authors.

Regarding the *resilience* (R) dimension, Q 5.0 is associated with the ability of production systems to adapt to crises, uncertainties, and changes [8]. From this perspective, the identified boosting elements prioritize the use of computerized predictive models and digital technologies to support adaptive decision-making in quality management, given the dynamic nature of industrial challenges [11, 24]. Also noteworthy are data-sharing practices through smart technologies along the supply chain, as well as the integration of active listening by consumers and other stakeholders [5, 25]. In crisis scenarios, the use of monitoring and analysis technologies strengthens organizational response capacity [11, 23, 26, 29, 36], enabling the detection and correction of failures and the rapid recovery of processes [19, 24, 32].

Some identified boosting elements operate across dimensions. In the integration between the *human-centricity* and *sustainability* (H/S) dimensions, it is observed that quality management expands its scope to incorporate economic, social, and environmental aspects, shifting the focus from exclusive customer satisfaction to the generation of value for society as a whole [11, 12, 22, 24, 26, 29–31, 33, 34]. In this context, Q 5.0 is understood as a management approach supported by digital systems that provide data to sustain sustainable innovation and strategic decision-making [30, 37, 42]. The widely recognized management models and standards, such as EFQM, ISO 9001, ISO 14001, and ISO 45001, remain structural benchmarks, serving as a foundation for the consolidation of Q 5.0 in organizations [12].

In the integration of the human-centricity and resilience (H/R) dimensions, the boosting elements highlight the role of quality managers as agents who promote integration between physical and digital systems, being responsible for integrating information, establishing standards, and conducting audits [30, 33, 37]. This integration reinforces organizational adaptability, favoring innovation trajectories supported by university–industry collaboration and R&D investments [37, 44]. Within this overlap between human-centricity and resilience, the development of training programs in innovation, supported by technologies such as AI and IoT, stands out, as they personalize learning, strengthen competencies, and expand managerial flexibility in the face of technological change [5, 11, 12, 19–27, 36]. The

integration of data into interoperable platforms and the use of adaptive systems help balance operational stability while keeping humans in the decision-making process [11, 24, 32, 34, 37]. Moreover, customizable digital platforms strengthen B2B and B2C relationships and promote higher-quality interactions [38]. Together, these initiatives support an organizational culture aligned with Q5.0 and change management [19, 20, 24, 26, 37].

Finally, some boosting elements of Q5.0 simultaneously encompass the human-centricity, sustainability, and resilience (H/S/R) dimensions. In this convergence, guidelines for developing quality management systems aligned with I5.0 stand out, focusing on improving working conditions and the centrality of the human being [12, 19, 24, 26–32]. These elements also integrate principles of sustainability and risk management, enhancing their capacity to respond to socioenvironmental and organizational challenges [16, 20, 24, 30, 34, 39]. Moreover, the use of technologies such as AI and BD is essential for identifying new customer demands and product customization, ensuring quality and considering sustainability and the generation of human and social value throughout the product lifecycle [5, 20, 24, 25, 34, 36–38].

4. Proposition and discussion of Quality 5.0 drivers

This section presents three proposed drivers for the development and improvement of Q5.0, namely, environmental sustainability and resilience; smart systems and human-machine collaboration; and social responsibility and quality of work life (Figure 5). These drivers are useful both for practical applications in different organizational contexts and for scientific exploration of the topic. They

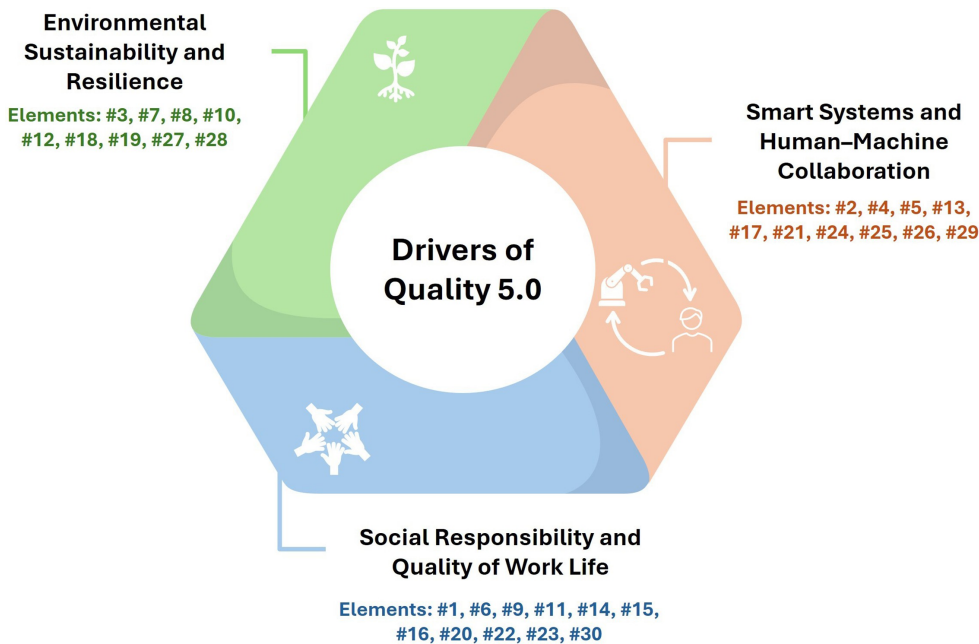


Figure 5.
Drivers for the development and improvement of Quality 5.0. Source: The authors.

were structured and grouped based on the predominant thematic affinity and strategic contribution of the 30 boosting elements of Q 5.0 (**Table 1**).

It is worth noting that, although each boosting element was assigned to the driver with which it has the strongest thematic affinity and predominant strategic contribution, several elements may also contribute to other drivers. Therefore, this assignment should not be interpreted as an exclusive relationship, since the proposed drivers are inherently interconnected and mutually reinforcing in the development and improvement of Q 5.0. Each driver is presented and discussed as follows, highlighting its boosting elements with the symbol #, followed by its numbering (#n).

4.1 Environmental sustainability and resilience

The environmental sustainability and resilience driver, in the context of Q 5.0, refers to the systemic integration of environmental sustainability into quality management systems, broadening the focus that has traditionally concentrated on customer satisfaction, to one that aims to create sustainable value for society (#3). Therefore, it is necessary to reorient quality management systems to be truly human-centered, integrating sustainability principles with risk management to anticipate vulnerabilities and support faster, more resilient decision-making (#8) [24, 30, 34].

Therefore, it is advisable to adopt I5.0-related technologies that aim to reduce waste and maximize energy efficiency to mitigate the potential negative environmental effects of productive activities (#18). At the same time, it is recommended that the supply chain be structured in line with resilience and risk management principles, using monitoring and data analysis technologies that enhance responsiveness and recovery in the face of critical events, operational disruptions, and crisis scenarios (#10). In addition, it is important to adopt and integrate standards-based management models and systems, such as EFQM, ISO 9001, ISO 14001, and ISO 45001, to establish organizational processes that are consistent, resilient, and aligned with the characteristics of Q 5.0 (#27) [24, 29, 37].

To structure data monitoring and analysis, it is recommended to enter economic, environmental, and social data into computerized management systems to support strategic decision-making, foster sustainable innovation, and promote a long-term systemic vision (#19). Thus, digital technologies will be used to support the development, monitoring, and analysis of quality management performance indicators from a sustainability and adaptive capacity perspective (#28) [11, 30, 37, 38, 42].

For a better transition to Q 5.0, it is also pertinent to implement concrete actions to gradually shape an organizational culture aligned with Q 5.0, including conducting pilot projects and planned quality change management actions, always considering the technical and technological limitations of each organization (#12). This movement should be supported by management and leadership models that encourage employee coparticipation and collaboration in digital environments, promoting knowledge exchange, collective learning, and the dissemination of successful practices (#7) [24, 26, 37, 39].

4.2 Smart systems and human-machine collaboration

Within the scope of Q 5.0, organizations should develop cyberphysical systems that integrate sensor data, computational models, and expertise to support

decision-making and ensure that workers remain primarily responsible for and in control of organizational processes (#4). To this end, data from automated processes should be integrated into interoperable platforms, thereby promoting intersectoral communication and enabling more accurate and timely human intervention (#13) [12, 32, 35].

In this regard, it is recommended to invest in equipment that enables human-machine collaboration, correcting failures in real time throughout the production process and the equipment life cycle to increase reliability, safety, and operational continuity (#17). In contexts of constant change, it is recommended to adopt computerized predictive models to identify problems and make decisions, thereby increasing the organization's agility in the face of uncertainties and new challenges (#25). This technological advancement needs to be incorporated into quality management systems, in line with the concepts of I5.0, particularly for optimizing working conditions, safety, and the valorization of human beings (#2) [24, 27, 31].

Concerning customers, it is essential to use technologies such as AI and BD to identify emerging demands and customize products, increasing efficiency, quality, and the generation of social, sustainable, and human value throughout the entire product cycle (#5). In addition, it is advisable to use AI to select and acquire robots for customer service and support, so that automation not only improves operational efficiency but also reinforces employee satisfaction, trust, and commitment (#21). Information coordination should be supported by the creation of adaptive databases that can harmonize operational stability with constant organizational learning (#24) [36, 43].

In addition, it is important to create customizable digital platforms that consider customer history, such as previous transactions, purchasing preferences, and interactions, strengthening B2B and B2C relationships and increasing transparency, trust, and satisfaction (#29). Finally, it is recommended to share data effectively throughout the supply chain while actively listening to consumers and other stakeholders to increase integration among chain links, improve decision-making, and serve as a strategic element of Q 5.0 (#26) [5, 25, 38].

4.3 Social responsibility and quality of work life

Q 5.0 presupposes the implementation of robotic systems and human-robot collaboration features, integrating online monitoring of performance indicators and human-centered factors such as safety, mental load, physical and cognitive ergonomics, trust, and acceptance, ensuring safe, efficient interactions focused on worker well-being (#15). This movement should be based on the responsible and ethical use of digital technologies, focusing on improving the employee experience and mitigating possible negative social impacts resulting from technological innovation (#6) [12, 22, 26, 41].

In Q 5.0, quality managers act as agents of integration between organizational performance and social demands, assuming responsibilities related to governance, standards-based management systems, and audit processes from a more humanized and strategic perspective (#20). This role underscores the need to drive technological innovation and efficiency to meet workers' needs, promote workplace well-being, and value human capital (#14) [30, 37, 39].

Consequently, the application of advanced technologies to complement employees' knowledge, creativity, and decision-making abilities significantly

improves the quality of products and services (#16). It should be noted that the participation of multiple stakeholders in defining clear, traceable, and accessible policies, strategies, and quality decisions reinforces shared responsibility and inclusion in organizational processes (#30) [12, 24, 30].

To support decision-making, employees need to participate in training programs focused on innovation and technologies such as AI and IoT to strengthen human skills and increase flexibility in the face of technological transformations and market demands. To this end, topics related to ethics, social responsibility, and the conscious use of technologies should be incorporated (#1) [26, 29, 31].

In this context, the integration of digital systems for continuous training to support sustainable innovation in industrial environments stands out, supported by university–industry collaboration and structured investments in research and development, thereby expanding the innovative and socially responsible capacity of organizations (#22). This organizational capacity can be fostered by incorporating social indicators into digital quality management systems. The indicators enable organizations to monitor social impacts and align their practices with the SDGs, including gender diversity, equity in leadership positions, and working conditions (#23) [37, 42, 44].

Digital platforms can also help promote the quality of life at work by enabling the systematic collection of employee and other stakeholder feedback, contributing to better alignment of production processes with human characteristics, advances in organizational culture, and improvements in occupational health and safety conditions (#11). The development of corporate health models based on digital systems enables the sharing of relevant data between workers, health professionals, and other stakeholders. This promotes organizational learning, workforce engagement, and decision-making that favors workers (#9) [21, 22, 31, 40].

5. Conclusion

Given the dynamism of social, economic, environmental, and technological transformations, quality management continues to evolve constantly to meet customer needs and ensure the survival of organizations. With the advent of I4.0, the use of digital technologies in companies intensified. I5.0 expanded on I4.0 by incorporating human values, environmental concerns, and a focus on adaptability and overcoming challenges. This led to the emergence of Quality 5.0 (Q5.0), which integrates advanced digital technologies into quality management, placing humans, socioenvironmental issues, and resilience at the center of organizational decision-making. Q5.0 represents an expansion and improvement of the quality models previously adopted by companies rather than their replacement.

This chapter proposes three drivers that support the development and improvement of Q5.0 in organizations. These drivers serve as a strategic guide for organizations to update themselves and successfully consolidate the integration of Q5.0. To this end, a SLR was conducted, revealing that despite its relevance and benefits, discussions of Q5.0 remain in their early stages, with few published works on the subject. Thus, this chapter contributes to the consolidation of Q5.0 in the literature by providing a theoretical framework that supports new research in the area. Moreover, it contributes in a practical, applied manner, supporting companies

in improving operational results, quality of life, and organizational resilience despite constant transformations.

Ultimately, it is worth highlighting that Q 5.0 is an expanding trend that is increasingly playing a strategic role for the future of organizations. However, the topic still requires further maturation, both theoretically and practically, as well as further advances in empirical investigations across different sectoral contexts. Thus, as the aim is to consolidate the link among technology, human values, socioenvironmental objectives, and resilience in quality management, Q 5.0 is increasingly becoming a new management logic, contributing to sustainable and human development in companies.

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